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M. Tech. (First Semester) Examination, Nov.-Dec. 2019

(Old Scheme)

(Elect. Engg. Branch)

HIGH VOLTAGE ENGINEERING & MEASUREMENT

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 40

Note *: All question are compulsory. Attempt any* ***two*** *parts from (a), (b), (c). All questions carries 10 marks.*

Unit-I

1. (a) Describe with neat diagrams the principle of operation application and limitations of Van De Graaf generation.

(b) Explain with neat diagram the principle of operation of :

(i) Series

(ii) Parallel resonant circuits for generation of high a.c. voltage. Compare their performance .

(c) A 12-stage impulse generator has capacitors each rated at 0·3 μF, 150 kV. The capacitance of the test specimen is 400 μF. Determine the wave front and wave tail resistance to produce at 1·2/50 μsec. impulse wave. Also determine the maximum output voltage if the charging voltage is 125 kV.

Unit-II

2. (a) Explain with neat diagram the principle of operation of an electrostatic voltmeter. Discuss its advantages and limitations for high voltage measurements.

(b) Draw chubb-fortesue circuit for measurement of peak value of a.c. voltage discuss its advantages over other methods.

(c) Discuss various resistance potential dividers and compare their performance of measurement of impulse voltage.

Unit-III

3. (a) Discuss the principle and construction diagram of Hall effect generator. Discuss its application and limitation.

(b) What are the different types of resistive shunt used for impulse current measurements? Discuss their characteristics and limitations.

(c) What is Rogowski coil? Explain with a neat diagram its principle of operation for measurement of high impulse currents. https://www.csvtuonline.com

Unit-IV

4. (a) What is Corona discharge? Explain clearly Anode and Cathode coronas.

(b) Explain Penning effect when referred to gaseous discharges.

(c) Explain streamer or kanal mechanism of spark for breakdown of gases.

Unit-V

5. (a) Explain briefly various theories of breakdown in liquid dielectrics.

(b) What do you mean by 'Intrinsic Strength' of a solid dielectric? Explain electric breakdown of solid dielectrics.

(c) What are 'Treeing' and 'Tracking'? Explain clearly the two processes in solid dielectric.